MILITARY SPECIFICATION

TEA MIX, INSTANT, SWEETENED, LEMON FLAVORED - (OPERATIONAL RATION COMPONENT)

This specification is approved for use by the Natick Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers a lemon flavored, sweetened, instant tea mix for preparation of a hot or cold beverage. The product is intended for use by the Department of Defense as a component of operational rations.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-P-44276 - Packaging of Ration, Cold Weather Components (In Pouches)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8955

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.2 Other Government documents. The following other Government documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.2 Other publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Application for copies should be addressed to the American Association of Cereal Chemists, 3340 Pilot Knob Road, St. Paul, MN 55121.)

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS (AOAC)

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

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NATIONAL ACADEMY OF SCIENCES

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4, 6.1, and 6.2).

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent and insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Sugar, white, granulated. Sugar shall be white, refined, granulated cane or beet sugar, or a combination thereof.

3.2.2 Tea, instant. Instant tea shall be a dry, 100 percent black (orange pekoe or pekoe) tea made by the extraction, concentration and dehydration of leaves from Thea sinesis. The instant tea shall be easily dissolved in hot or cold water (see 6.3).

3.2.3 Citric acid, anhydrous, fine granulated. Anhydrous fine granulated citric acid shall comply with the requirements of the Food Chemicals Codex.

3.2.4 Lemon juice powder. Lemon juice powder shall be equal to Beatreme 3049 manufactured by Beatrice Food Ingredients, Inc.

3.2.5 Magnesium oxide powder. Magnesium oxide powder shall be food grade and shall comply with the requirements of the Food Chemicals Codex. The powder shall contain a minimum of 96 percent magnesium oxide.

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3.3 Preparation and processing. Processing shall be on a continuous basis.

3.3.1 Formulation of the instant tea mix. The instant tea mix shall be formulated as follows:

Ingredients	Percent by Weight
Sugar, white, granulated	89.0
Tea, instant	4.2
Citric acid	4.0
Lemon juice powder	2.5
Magnesium oxide powder	0.3
	100.0

3.3.2 Preparation of instant tea mix. All ingredients shall be mixed thoroughly in a Hobart or other suitable mixer. Alternate methods of production to reduce ingredient separation may be used to meet finished product requirements (see 3.5).

3.4 Pouch filling and sealing. The product shall be filled and sealed in pouches such as to meet the requirements in 3.5 and MIL-P-44276.

3.5 Finished product requirements. The finished product shall comply with the following requirements:

a. There shall be no foreign material such as, but not limited to, dirt, insect parts, hair, wood, glass, or metal.

b. There shall be no foreign odor or flavor such as, but not limited to, burnt, scorched, stale, sour, rancid, or moldy.

c. There shall be no color foreign to the product.

d. The contents of each pouch shall weigh not less than 14.0 grams and not greater than 17.0 grams.

e. The powder shall be white crystals with dark brown aggregates or agglomerated light brown crystals.

f. The product shall be a uniformly blended, free-flowing mixture.

g. The product shall not contain lumps which do not fall apart under light finger pressure.

h. The moisture content of the product shall not exceed 1.0 percent.

i. When product is rehydrated as specified in footnote 3 of table I, the product shall possess flavor, odor, and texture similar to that of high-quality instant tea, sweetened and flavored with lemon.

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3.5.1 Palatability. The finished product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

3.6 Plant qualification. The product shall be prepared, processed, and packaged in establishments meeting the requirements of Title 21, Code of Federal Regulations, Part 110: Current Good Manufacturing Practice in Manufacture, Processing, Packaging or Holding of Human Foods; and the plant sanitation requirements of the appropriate Government inspection agency.

3.7 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this document. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of inspection necessary to assure compliance with the requirements of this document.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this document and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this document or any appearance or palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

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4.5.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, or compliance be verified by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods, such as the standard methods described in the Official Methods of Analysis of the Association of Official Analytical Chemists and in the Approved Methods of the American Association of Cereal Chemists, to determine conformance to the condition requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, and filling requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved product.

4.5.3 Net weight inspection. The net weight of the filled and sealed pouches shall be determined by weighing each sample unit on a suitable scale tared with a representative empty pouch. Results shall be reported to the nearest 0.1 gram. Any individual net weight of less than 14.0 grams or more than 17.0 grams shall be scored as a major defect. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-3 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5.

4.5.4 Product inspection. The finished product shall be examined for the defects listed in table I. The lot size shall be expressed in pouches. The sample unit shall be the contents of one filled and sealed pouch. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 1.5.

TABLE I. Product defects 1/ 2/

Category	Defect
Major	Dry product
101	Not white crystals with dark brown aggregates or agglomerated light brown crystals
102	Not a uniformly blended, free-flowing mixture
103	Presence of lumps which do not fall apart with light finger pressure

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TABLE I. Product defects 1/ 2/ (cont'd)

Category	Defect
Major	Prepared beverage 3/
104	Product does not possess flavor, odor, and texture similar to that of a high quality instant tea sweetened and flavored with lemon.

1/ The presence of foreign material (e.g. glass, dirt, insect parts, hair, wood, metal), foreign odor or flavor (e.g. burnt, scorched, moldy, rancid, sour, stale), or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by an Agricultural Marketing Service (AMS) agent.)

3/ Add 8 ounces of water to contents of one packet of tea and stir. One half of the samples shall be reconstituted in hot water and the other half in cold water.

4.5.5 Moisture content testing. The moisture content of the dry product shall be determined in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists (AOAC); Chapter: Sugar and Sugar Products; Section: Moisture; Method: Drying in Vacuum. The results shall be reported to the nearest 0.1 percent. Any result not conforming to the requirement in 3.5 shall be classified as a major defect. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 1.5.

4.5.6 Packaging inspection. Inspection of packaging shall be in accordance with the quality assurance provisions of MIL-P-44276.

5. PACKAGING

5.1 Packing. Packing shall be in accordance with the requirements of MIL-P-44276.

6. NOTES

6.1 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. When a first article is required (see 3.1, 4.4, and 6.2).
c. Provisions for approved preproduction samples (see 3.5.1 and 6.2).

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6.2 First article. When a first article sample is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.3 Tea, instant. It has been found that instant tea manufactured by Nestea meets the requirements of 3.2.2 and performs satisfactorily in this product.

6.4 Subject term (key word) listing.

Beverage
Cold weather rations
Operational ration component
Ration
Tea

Custodian:

Army - GL

Review activity:

Army - MD, TS
Navy - MC, SA
Air Force - 50
DP - SS

Preparing activity:

Army - GL

Project No. 8955-0057

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-44284(GL)	Tea Mix, Instant, Sweetened, Lemon Flavored - (Operational Ration Component)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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